Final Project: Wine Prices and Regional Price Premia

Instructions and deliverables

This take-home project is the final assessment. Work in the assigned groups and submit two files before the presentation date. Then, you will present your work and answer questions from the lecturer.

Deliverable	What it must contain
Excel workbook	One sheet per part (A, B, C). Every statistic, test and regression you report must be computed in this workbook, using formulas or the Data Analysis ToolPak.
Short report (Word or PDF, at most [8] pages)	Answers in question order (A1, A2, …). For each answer: the key number or table copied from Excel, then 2–4 sentences of interpretation.

Name both files after the group name, e.g. Group_1.xlsx and Group_1.docx, and list every member on the first page of the report, with their respective contribution (in detail tasks and percentage of total contribution).

Rules for every hypothesis test:

1. State H_0 and H_1.
2. Name the test statistic and the Excel output or formula it comes from.
3. Give the p-value (or the critical value) at $\alpha = 0.05$.
4. State the decision and what it means in the wine context.

You may discuss the project only with your group members and the instructor. Remember to show the formula in the Excel file, raw Excel output pasted without interpretation earns no credit.

The case

An association of Napa Valley wineries has hired you to answer one question: do buyers pay more for a bottle labelled “Napa Valley”, and if so, how many dollars more?

In the US, certified wine regions called AVAs (American Viticultural Areas) may print their name on the label. The association also wants to know whether its main competitor, Sonoma, earns a price premium, and whether that premium is larger or smaller than Napa’s.

You will answer with a hedonic price model: the price of a wine is explained by its characteristics (quality score, age, production volume, region, grape variety). In such a model, each coefficient is the implicit price of one characteristic, holding the others constant.

The final project follows the following steps, from a single mean to a full regression, before you answer the final association’s question.

$$\text{Price} = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \varepsilon$$

The dataset

The file data contains 806 California wines of the 2001 vintage, taken from Wine Spectator tasting records. Each row is one wine.

Variable	Type	Meaning
Price	Continuous	Retail price per bottle, in US$
Cases	Continuous	Number of cases produced, in thousands (30 = 30,000 cases)
Score	Continuous	Wine Spectator quality score from a blind tasting (out of 100)
Age	Continuous	Years of ageing before the wine was sold
Region Dummy: Napa valley, bay area, sonoma, sierra/foothills, mendocino / lake	Dummy (0/1)	Region where the wine was produced
Variety Dummy: Nonvarietal, Pinot noir, Cabernet, Merlot, Shyrah	Dummy (0/1)	Grape variety; Nonvarietal = a blend of several grapes

Two dummies are deliberately missing. A wine with 0 in all five region dummies is a Generic California wine (no AVA). A wine with 0 in all five variety dummies is implicitly understood as a Zinfandel.

Part A · Data and description (20 points)

Part A covers steps 1–3 of the analysis process: the question, describing the data, and visualizing it

A1 (3 points). Besides measuring the Napa premium price compared to other regions, name two other people or organizations who could use this model, and what each would use it for.

A2 (3 points). For Cases, Score and Age, and for each set of dummies (region, variety), state the sign you expect on the coefficient and explain why in one sentence.

A3 (4 points). Build one table of summary statistics for Price, Cases, Score and Age: mean, median, standard deviation, variance, minimum, maximum and skewness. Compare the mean and the median of Price. What does the gap tell you about its distribution?

A4 (4 points). Draw three scatter plots: Price against Cases, Price against Score, and Price against Age. For each, say whether the relationship looks positive or negative, whether it looks like a straight line, and whether any points stand far from the rest.

A5 (3 points). Draw a histogram of Price and a histogram of ln(Price). Which one looks closer to a normal distribution? Which version of Y would you rather put on the left of a regression, and why?

A6 (3 points). Normality of Y is often given as a condition for valid t-tests on regression coefficients. With 806 observations, should you worry a lot if Price is not normal?

Part B · The Modeling (45 points)

B1 · Continuous Y, no X (5 points). Compute the mean of Price and its 95% confidence interval. Test H_0: $\mu = 25$ against H_1: $\mu \neq 25$. Then explain in one sentence why this mean is the estimate of β_0 in the model Price = $\beta_0 + \varepsilon$.

B2 · Continuous Y, one dummy X (8 points).

1. Compare the mean Price of Napa wines with all other wines using the Excel t-test for two samples with unequal variances (Welch). State the hypotheses and the decision.
2. Regress Price on the Napa valley dummy alone. Show that the slope equals the difference between the two group means.
3. The t statistic of the slope is not the same as the Welch t. Which two-sample t-test does the regression reproduce exactly, and why?
4. Given your histogram in A5, which rank-based test from class would be a safer check here? You do not need to run it.

B3 · Continuous Y, one X with 3+ groups (8 points).

1. Create a Region column with six groups (the five AVAs plus Generic California). Run a one-way ANOVA of ln(Price) across the six regions and state the decision.
2. Regress ln(Price) on the five region dummies. Show that SSR, SSE and F match the ANOVA table from step 1.
3. Why must one region dummy be left out of the regression? Interpret the Napa valley coefficient.
4. If Sonoma were the left-out region instead of Generic California, what would the Napa valley coefficient measure, and what value would it take? Answer from your step 2 output, without re-running.

B4 · Continuous Y, one continuous X (6 points).

1. Compute the correlation between Price and Score, and between ln(Price) and Score. Which is stronger, and why might that be?
2. Regress ln(Price) on Score. Interpret the slope in percent per Score point.

B5 · Continuous Y, several X (12 points).

1. Linear model: regress Price on all 13 explanatory variables. Comment on which variables are significant, on R^2, and on the overall F-test.
2. Log-linear model: regress ln(Price) on the same 13 variables. Report R^2 and the F-test. Interpret the coefficients of Cases, Score and Age, one region coefficient and one variety coefficient.
3. For each model, plot residuals (y axis) against predicted values. Compare the two plots: what did the log transformation fix?
4. The Napa valley coefficient in the log model differs from the one in B3. Explain why, using the idea of controlling for other variables. Which variable do you think matters most for this change?
5. Name one variable not in the dataset that could bias the Napa coefficient if left out, and say in which direction.

B6 · Categorical Y, categorical X (6 points).

1. Build a crosstab of Region (Napa valley, Sonoma, Other) against Variety (Cabernet, Merlot, Pinot noir, Other). Add row percentages.

2. Run Pearson's χ^2 test of independence. Compute the expected counts in Excel, check that all are at least 5, and report χ^2, degrees of freedom and the p-value (CHISQ.TEST or CHISQ.DIST.RT).
3. What does the result mean for the simple Napa-versus-others comparison in B2? Link your answer to B5.4.

Part C · Conclusions for the Napa association (25 points)

Choose the model from Part B you trust most, and use it to answer the client. Remember that there is no single correct model. Credit goes to a well-argued choice and to reading every coefficient in the way that matches the model you chose.

C1 · Choose your model (4 points). Choose either the linear model (B5.1) or the log-linear model (B5.2) as the basis for your answers. Justify the choice with at least two pieces of evidence from Parts A and B, for example the distribution of Price, the fit of the two models, the residual plots or the significance of the region dummies. State the reference region and the reference variety of your model.

C2 · Napa premium (5 points). Using your chosen model, state the price premium of the Napa Valley name compared with Generic California, other things equal, as a dollar amount per bottle, and say whether it is statistically significant. If your model gives the premium in another unit first, show how you converted it to dollars, which price level you used, and why that price level is a sensible choice.

C3 · Napa versus Sonoma (5 points). Compute the Sonoma premium in the same way. Then re-estimate your chosen model with Sonoma as the left-out region (drop the sonoma dummy, add a Generic California dummy). Is Napa's premium significantly larger than Sonoma's? Report the relevant coefficient, its p-value and your decision.

C4 · Significant is not important (4 points). Using your chosen model, compute the price change from producing 10 more units of Cases (10,000 cases), from one more Score point, and from one more year of Age, in the unit your model gives. Is the Cases effect large enough to matter for a winery's pricing? Compare its size with its p-value, and contrast it with a coefficient that is not significant in your model.

C5 · Premium or quality? (4 points). Does a price premium for Napa prove that Napa wines are of higher quality? Look at which variables your model holds constant, and state precisely what the Napa coefficient measures.

C6 · Message to the client (3 points). Write one paragraph of at most 150 words for the association's board, with no statistical jargon: is there a Napa premium, how many dollars per bottle, how does it compare with Sonoma, and what are the limits of this analysis?

Grading

The project is worth 100 points.

Part	Content	Points
A	Data and description	15
B	Building the model	35
C	Conclusions for the client	20
Report quality	Clear structure, every number traceable to the Excel workbook, complete test steps, sentences rather than raw output	5
Individual Q&A	Answering questions	25
Total		100